



NORTHERN UNIVERSITY
OF BUSINESS & TECHNOLOGY
K H U L N A

Backend Progress Report for AI Prompt Engineering Learning Hub

Course Code: CSE4104

Course Title: Software Development III

Team: CSE4104-7C-T02

Department of Computer Science and Engineering

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Team Information Sheet

Team Name:	CSE4104-7C-T02
Section:	7C
Project Title:	AI Prompt Engineering Learning Hub
GitHub Repository Link:	https://github.com/RahminHossain/cse41047c-t02-ai-prompt-engineering-learning-hub
Team Leader:	Rahmin Hossain ID: 11230121156

Team Members Roles & Credentials

Sl	Members Name	Student ID	Role/Responsibility
1	Lamiha Faija	11230121140	Frontend Developer & UI Researcher
2	Fatema Akter Elma	11230121142	AI/Prompt Integrator
3	Masum Billah	11230121149	Backend Developer & QA Tester
4	Rahmin Hossain	11230121156	Database Admin & Data Engineer

Project Title: AI Prompt Engineering Learning Hub

1. Backend Technology Stack

<u>Component</u>	<u>Technology</u>
Runtime Environment	Node.js
Backend Framework	Express.js
Database	MongoDB Atlas
ODM	Mongoose
Authentication	JWT (JSON Web Token)
Password Security	bcrypt

Configuration & Setup:

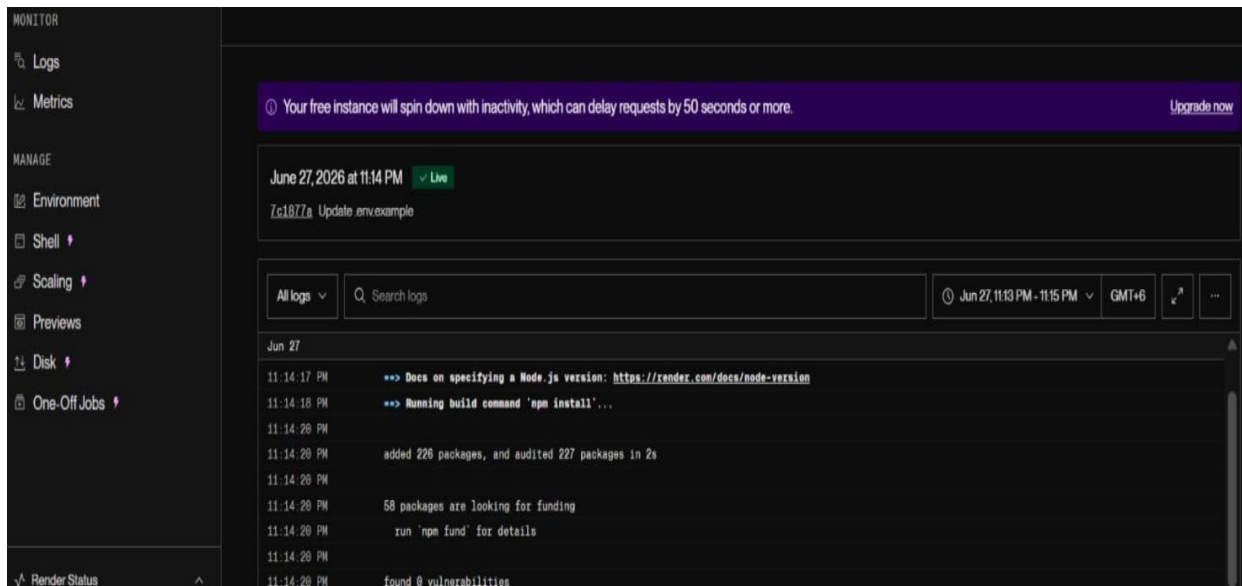
The backend was successfully configured for production deployment using Node.js, Express, and MongoDB Atlas. package.json Scripts The build and start scripts were configured for Render deployment:

```
"scripts": {  
  "dev": "nodemon src/server.js",  
  "start": "node src/server.js",  
  "lint": "eslint .",  
  "test": "node --test",  
  "seed": "node seed/seed.js"  
}
```

Render Deployment Status:

The application was deployed via a Render Web Service with the following settings:

- Root Directory: *backend*
- Build Command: *npm install*
- Start Command: *npm start*



The screenshot displays the Render deployment status interface. On the left is a sidebar with navigation options: MONITOR (Logs, Metrics), MANAGE (Environment, Shell, Scaling, Previews, Disk, One-Off Jobs), and Render Status. The main panel shows a deployment for 'June 27, 2026 at 11:14 PM' with a 'Live' status. Below this is a search bar for logs and a log viewer showing the following output:

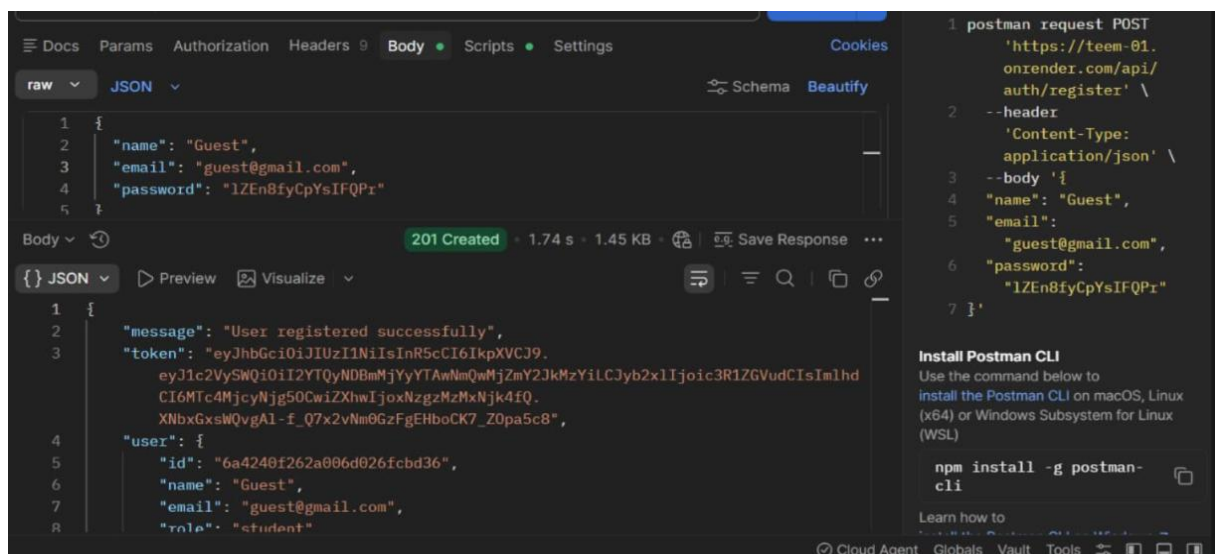
```
Jun 27
11:14:17 PM    ==> Docs on specifying a Node.js version: https://render.com/docs/node-version
11:14:18 PM    ==> Running build command 'npm install'...
11:14:20 PM
11:14:20 PM    added 220 packages, and audited 227 packages in 2s
11:14:20 PM
11:14:20 PM    50 packages are looking for funding
11:14:20 PM      run 'npm fund' for details
11:14:20 PM
11:14:20 PM    found 0 vulnerabilities
```

API Endpoint Testing (Postman):

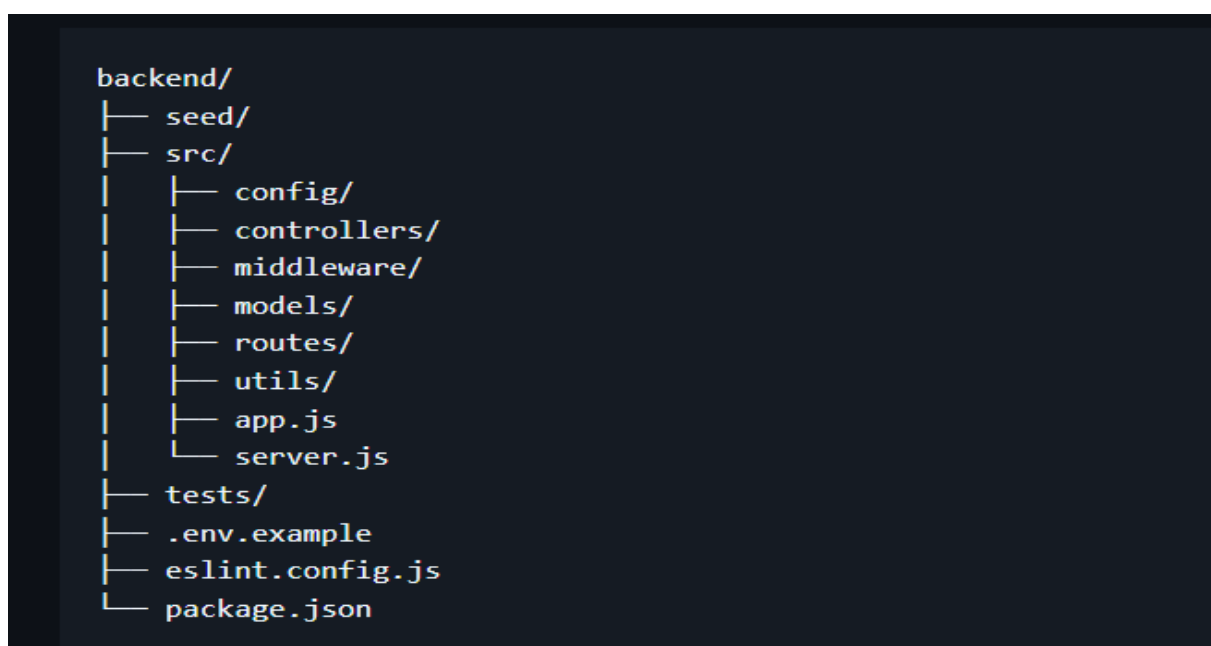
The deployed live API was successfully tested using Postman to verify database connectivity and routing.

Test Details

- Method: *POST*
- Live Endpoint: *https://teem-01.onrender.com/api/auth/register*
- Content-Type: *application/json*



Backend Folder Structure:



2. Database Design Summary (MongoDB)

Collections

users

- `_id` (ObjectId, primary key)
- `name` (String, required)
- `email` (String, required, unique)
- `password` (String, required, bcrypt hash)
- `role` (Enum: student, admin)
- `bio` (String)
- `createdAt`, `updatedAt`

learningresources

- `_id` (ObjectId, primary key)
- `title` (String, required)
- `description` (String, required)
- `level` (Enum: beginner, intermediate, advanced)
- `category` (String, required)
- `createdBy` (ObjectId, ref: users)
- `isPublished` (Boolean)
- `createdAt`, `updatedAt`

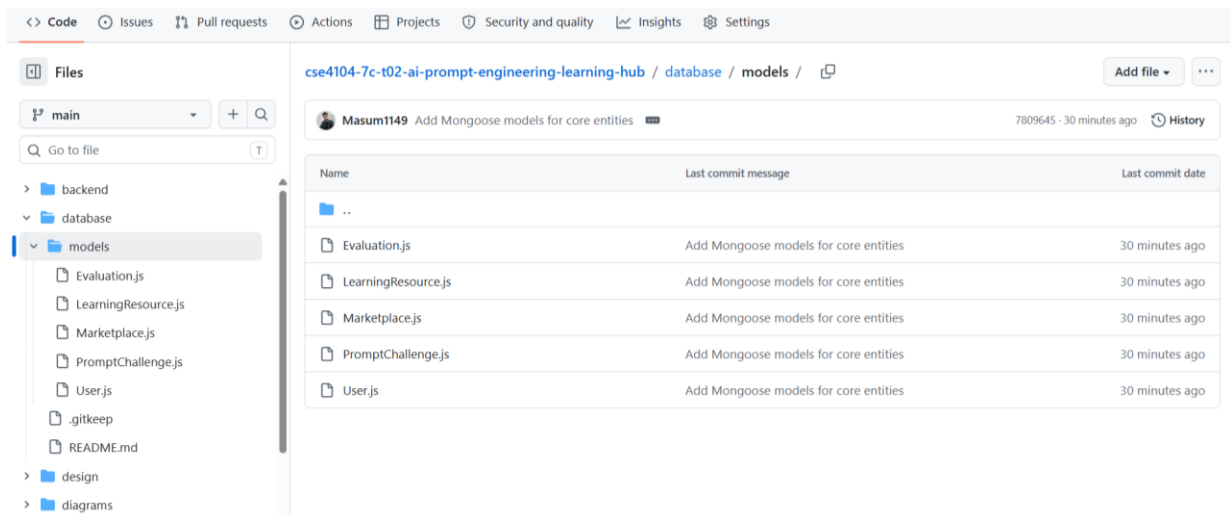
promptchallenges

- `_id` (ObjectId, primary key)
- `title` (String, required)
- `promptTask` (String, required)
- `expectedOutcome` (String, required)
- `difficulty` (Enum: easy, medium, hard)
- `createdBy` (ObjectId, ref: users)
- `createdAt`, `updatedAt`

Relationships

- A user can create many learning resources.
- A user can create many prompt challenges.
- createdBy is a foreign key reference from learning resources and challenges to users.

Database Collection:



3. Implemented APIs:

Authentication APIs

- POST /api/auth/register
- POST /api/auth/login
- POST /api/auth/logout

User APIs

- GET /api/users/profile
- PUT /api/users/profile

Learning Resources APIs

- GET /api/resources
- GET /api/resources/:id

- POST /api/resources (admin)
- PUT /api/resources/:id (admin)
- DELETE /api/resources/:id (admin)

Prompt Challenges

- GET /api/challenges
- GET /api/challenges/:id
- POST /api/challenges (admin)
- PUT /api/challenges/:id (admin)
- DELETE /api/challenges/:id (admin)

4. Authentication Workflow:

The authentication system uses JWT (JSON Web Token) for secure user authentication.

Workflow

1. User registers using Full Name, Email, and Password.
2. The password is securely encrypted using bcrypt before being stored in MongoDB.
3. The user logs in with a registered Email and Password.
4. The server validates the user credentials.
5. Upon successful authentication, a JWT token is generated.
6. The client stores the JWT token securely.
7. The JWT token is included in the Authorization header for protected API requests.
8. The backend verifies the token before granting access to protected resources.

5. Current Development Progress:

Completed

- Project setup and folder structure
- Express.js server configuration
- MongoDB Atlas connection
- Mongoose schema design
- JWT Authentication
- Password hashing with bcrypt
- User Registration and Login APIs
- User Profile APIs
- Learning Module APIs
- Prompt Playground APIs
- Google Gemini AI integration
- AI Evaluation APIs
- Challenge APIs
- API testing using Postman
- Global error handling

In Progress

- Performance optimization
- Backend debugging

Remaining

- Frontend integration
- End-to-end testing
- Cloud deployment
- Final project demonstration

6. Github Repository Link: <https://github.com/RahminHossain/cse4104-7c-t02-ai-prompt-engineering-learning-hub>